

HR / POWER

Report di identificazione

Run 97cbc531c72a4f12bce004944cb521da

Dataset: Pre-window

Creazione: 2026-10-05T16:19:42.058263+00:00 | Stato: completed

Struttura: short_transient | $dx/dt = \gamma (P(t-L)-P_0)$; $HR_{hat} = HR_0 + x$

Inizializzazione: equilibrium | $P_0 = \text{mean}(P_{pre})$, $HR_0 = \text{mean}(HR_{pre})$; $x(0)=0$; $P(t<0)=P_0$

Dataset: 1 segmenti; train richiesti: 1; risultati: 1; validi: 1; falliti: 0.

Ambito identificazione: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

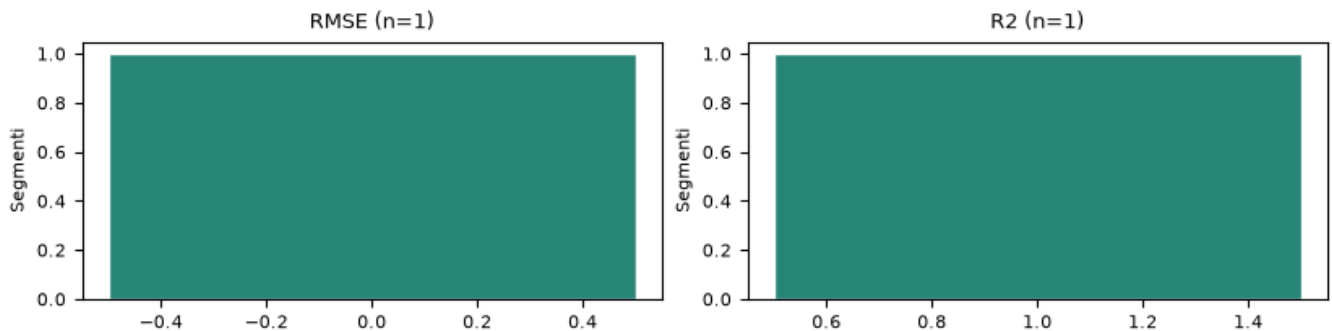
Pre-window richiesta: si; durata: 10 s. P_0 : pre_window_mean; HR_0 : pre_window_mean

Pre-window disponibili: 1 / 1 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R^2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

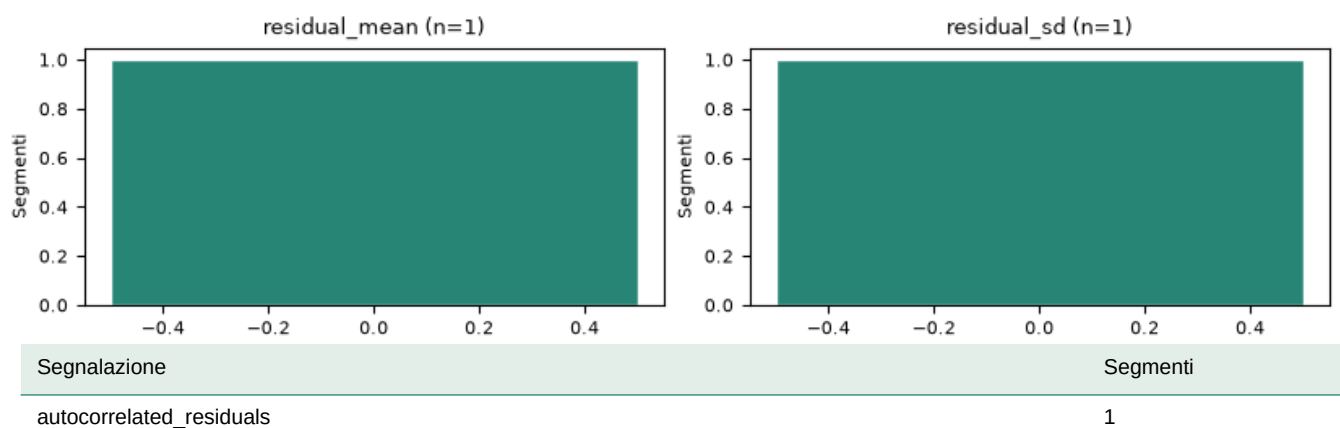
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	1	6.1492e-11	6.1492e-11	n.d.	6.1492e-11	6.1492e-11
MAE	1	4.7617e-11	4.7617e-11	n.d.	4.7617e-11	4.7617e-11
bias	1	2.7288e-11	2.7288e-11	n.d.	2.7288e-11	2.7288e-11
R^2	1	1	1	n.d.	1	1
amplitude_ratio	1	1	1	n.d.	1	1



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	1	2.7288e-11	2.7288e-11	n.d.	2.7288e-11	2.7288e-11
residual_sd	1	5.6467e-11	5.6467e-11	n.d.	5.6467e-11	5.6467e-11
residual_autocorrelation_lag1	1	0.78533	0.78533	n.d.	0.78533	0.78533
residual_autocorrelation_lag5	1	0.11848	0.11848	n.d.	0.11848	0.11848



Multistart

Ricerca: multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 2; convergenti: 2. Accordo di tutti gli start: 1 / 1 fit validi.

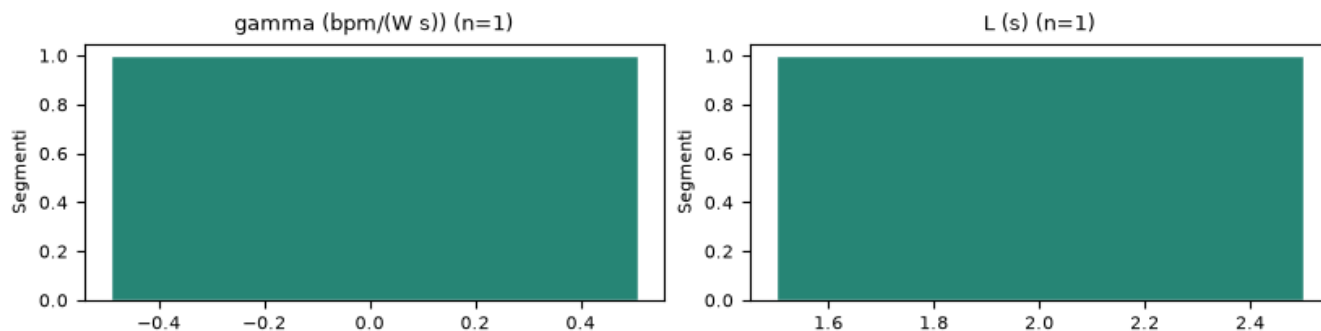
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	2	1.9043e-19	1.9043e-19	1.5702e-19	7.9408e-20	3.0146e-19
nfev	2	17	17	9.8995	10	24

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
gamma (bpm/(W s))	1	0.008	0.008	n.d.	0.008	0.008
L (s)	1	2	2	n.d.	2	2

Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
gamma	2.8718e-14	3.5898e-10	0	0
L	3.859e-11	1.9295e-09	0	0



Identificabilita

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unita di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.

- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualita della traiettoria.

- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.

- Multistart: SSE simili con parametri differenti indicano ambiguita. CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	0	n.d.	n.d.	n.d.	n.d.	n.d.
correlation_K_tau	0	n.d.	n.d.	n.d.	n.d.	n.d.
Jacobian rank	1	2	2	n.d.	2	2
Jacobian condition	1	2741.3	2741.3	n.d.	2741.3	2741.3

ρ non applicabile alla struttura gamma, L. K e τ non vengono ricostruiti.

Scheda segmento

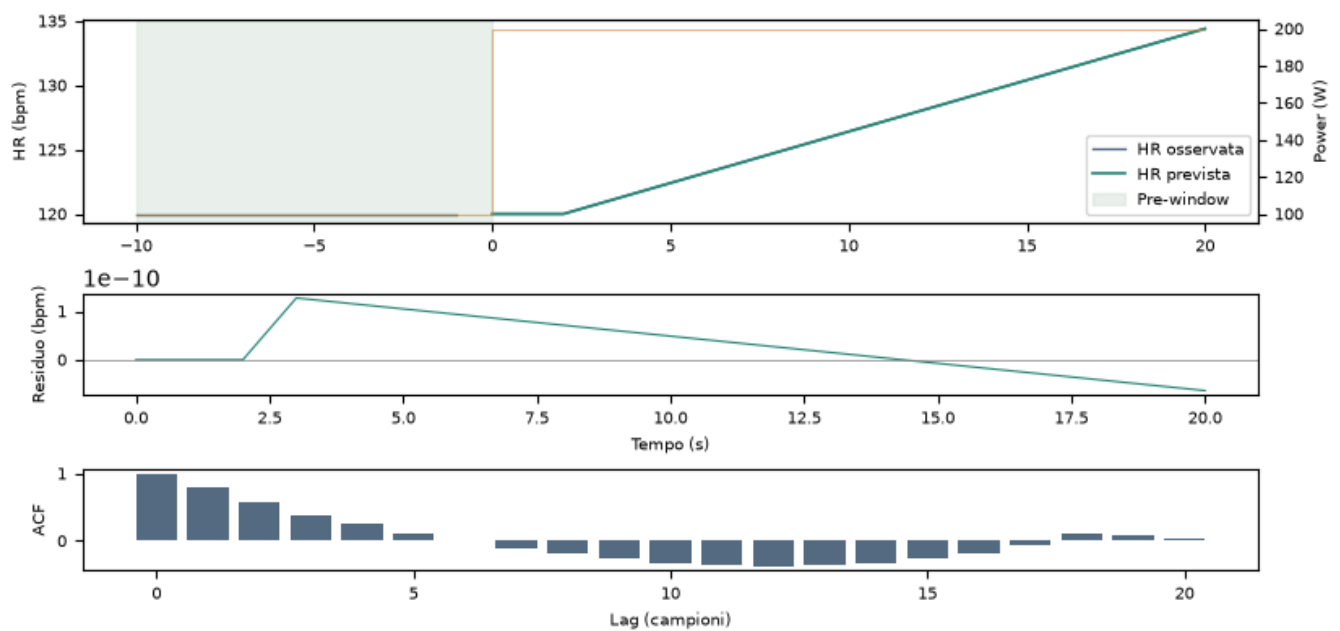
a.csv

ID: 3d7f2fd67f5810f5c2483159 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=100 W; HR0=120 bpm.

RMSE	MAE	bias	R2	amplitude_ratio
6.1492e-11	4.7617e-11	2.7288e-11	1	1
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
2.7288e-11	5.6467e-11	0.78533	0.11848	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
gamma	0.008	2.8718e-14	0.008 / 0.008	3.5898e-10	no	no
L	2	3.859e-11	2 / 2	1.9295e-09	no	no

rho=n.d.; Jacobian rank=2/2; cond=2741.3; accordo multistart=si.

Correlazioni: {"gamma_L": 0.8716185636684902}

Multistart: 2 tentativi, 2 convergenti; SSE min/max: 7.9408e-20 / 3.0146e-19

Segnalazioni: autocorrelated_residuals

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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